

Software Requirements Specification (SRS): AI & LMS-Enabled Cooperative Capacity Building, ERP, and Employment Ecosystem

Document Control and Metadata

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Executive Summary and Source Problem Statement

The National Council for Cooperative Training (NCCT), functioning as an autonomous society under the Ministry of Cooperation, Government of India, oversees executive education,

management development, and grassroots operational training across India's cooperative sector¹. Operating through an apex management institute—the Vaikunth Mehta National Institute of Cooperative Management (VAMNICOM) at Pune—alongside 5 Regional Institutes of Cooperative Management (RICMs), 14 state-level Institutes of Cooperative Management (ICMs), and supporting approximately 109 Junior Cooperative Training Centres (JCTCs), the NCCT conducts more than 3,200 training programs annually for over 200,000 participants³. Beneficiaries span personnel from Primary Agricultural Credit Societies (PACS), District Central Cooperative Banks (DCCBs), State Cooperative Agriculture and Rural Development Banks (SCARDBs), dairy and fisheries cooperatives, Self-Help Groups (SHGs), and rural youth³. The official Problem Statement ID 26087 articulates that legacy operational workflows across NCCT institutions are largely manual or fragmented¹. Program scheduling, society nominations, batch formation, attendance capture, and logistics management frequently rely on disparate spreadsheets or isolated legacy portals¹. These disconnected practices result in administrative duplication, inadequate trainee tracking, limited learning analytics, poor employment linkages, and unverifiable paper-based certifications¹. Simultaneously, trained rural youth struggle to access formal employment, entrepreneurship linkages, and verifiable skill visibility, even as rural credit institutions and agro-enterprises actively seek certified personnel¹. This document specifies the software, hardware, cryptographic, and operational requirements for an integrated, cloud-native, and offline-capable digital ecosystem¹. The platform unites an Enterprise Resource Planning (ERP) training management engine, an offline-first multilingual Learning Management System (LMS), edge biometric attendance hardware, an Indic conversational AI career counselor, and a semantic candidate-job exchange linked to India's national Digital Public Infrastructure (DPI)¹.

Purpose, Scope, and System Boundaries

Purpose

The primary purpose of this platform is to provide a single digital operating system for cooperative capacity building in India¹. The platform standardizes the learner lifecycle across all twenty primary NCCT institutions and affiliated centers: program discovery, organizational nomination, campus logistics, edge identity verification, interactive multilingual instruction, digital assessments, W3C-compliant credential issuance, career guidance, and formal recruitment placement¹.

System Scope

The ecosystem encompasses:

- A centralized multi-tenant ERP platform for academic calendars, participant nominations, batch scheduling, classroom allocations, and hostel/mess logistics¹.
- An edge-compatible attendance system combining offline facial verification with dynamic, geofenced, Time-Based One-Time Password (TOTP) QR codes¹.
- An offline-first, SCORM- and xAPI-compliant LMS featuring automated Indic language translation and voice synthesis powered by MeitY's Bhashini Universal Language Contribution APIs (ULCA)¹.

- A cryptographically signed digital credentialing engine integrated with DigiLocker and API Setu using W3C Verifiable Credentials Data Model 2.0¹.
- An Indic voice-and-text career counseling chatbot utilizing Retrieval-Augmented Generation (RAG) grounded in official cooperative bylaws and training catalogs¹.
- An employer and recruiter exchange featuring multi-factor semantic candidate-job matching and bidirectional data bridges with the National Career Service (NCS) and Skill India Digital Hub (SIDH)¹.
- Management Information Systems (MIS) analytics dashboards providing operational oversight to the Ministry of Cooperation and NCCT headquarters¹.

Operational Boundaries and Out-of-Scope Items

- The platform does not serve as a transactional Core Banking System (CBS) for cooperative banks or as an inventory transaction engine for computerized PACS¹⁰. It interfaces with these systems strictly for employee nomination and skill verification¹.
- The platform does not require proprietary, custom-manufactured attendance hardware; all edge capabilities are designed for commercial off-the-shelf (COTS) devices (Android tablets, smartphones, and webcams)¹.
- Direct financial disbursements, stipends, or allowances are processed through external Public Financial Management System (PFMS) gateways rather than native digital payment processing¹.

Stakeholders and Comprehensive User Persona Matrix

Stakeholder Role	Access Tier & Scope	Core System Responsibilities & Typical User Journeys
Trainee / Rural Youth [cite: 1]	Tier 1: Client Application (Mobile / PWA) ¹ .	Discovers training courses; self-registers; completes offline micro-learning; marks edge biometric/QR attendance; takes assessments; downloads DigiLocker credentials; queries AI career bot; applies for cooperative jobs ¹ .
Cooperative Society Admin (PACS / DCCB / Milk Union) [cite: 1, 3, 6]	Tier 2: Sponsoring Organization Portal ¹ .	Submits bulk employee nominations; uploads sponsorship documentation; tracks staff

		completion metrics; reviews institutional training histories ¹ .
Training Coordinator (RICM / ICM Admin) [cite: 1, 3]	Tier 3: Institutional Campus ERP ¹ .	Configures course offerings; reviews nominations; forms cohorts; allocates faculty, classrooms, and hostel rooms; audits daily attendance logs; approves certificate generation ¹ .
Faculty / Resource Person [cite: 1]	Tier 3: LMS & Instructional Console ¹ .	Curates multilingual lessons; conducts live sessions; displays dynamic session QR codes; evaluates assessments; monitors learner progress; handles attendance overrides ¹ .
Campus Logistics / Hostel Warden [cite: 1]	Tier 3: Logistics & Facilities Console ¹ .	Manages bed inventory across campus hostels; oversees dining hall attendance and mess billing; tracks room vacancies and facility maintenance ¹ .
Recruiter / Cooperative Employer [cite: 1]	Tier 4: Employment Exchange Portal ¹ .	Posts job requisitions; configures required competency profiles; searches candidate databases; inspects cryptographically verified credentials; schedules interviews ¹ .
Institute Director (VAMNICOM / RICM / ICM)	Tier 5: Institutional Executive Dashboard ¹ .	Reviews campus-wide training targets, seat occupancy, hostel capacity,

[cite: 1, 3]		faculty workloads, and local budget utilization ¹ .
Central NCCT / Ministry Administrator [cite: 1]	Tier 6: Apex National Governance Portal ¹ .	Sets national training targets; monitors state-wide PACS computerization training progress; reviews macro-level employment outcomes; audits platform security ¹ .
System Administrator & Security Officer [cite: 1]	Tier 7: Root Platform Operations ¹ .	Manages role-based access control (RBAC), API Setu keys, HSM cryptographic keypairs, database failover protocols, and DPDP audit logs ¹ .

High-Level System Architecture and Multi-Tier Topology

The ecosystem uses a cloud-native microservices architecture, paired with local edge nodes deployed at training facilities to maintain operational continuity under limited wide-area network (WAN) connectivity¹.

- **Client & Edge Hardware Tier:** Trainees interface with the system via a cross-platform Flutter application or a lightweight Progressive Web Application (PWA)¹. The local client persists instructional media, lesson progress, and encrypted attendance logs within an embedded SQLite database using the Room abstraction layer¹. Classroom attendance is captured using wall-mounted COTS Android tablets or webcams running edge-optimized deep learning models (MobileFaceNet with ArcFace margin loss), operating completely offline¹.
- **API Gateway & Ingress Tier:** An enterprise API Gateway terminates incoming TLS 1.3 connections, applies rate-limiting rules via Redis, enforces Web Application Firewall (WAF) filtering, and handles JSON Web Token (JWT) verification¹.
- **Core Application Services Mesh:** Containerized services run in an autoscaling Kubernetes cluster. Independent services govern Program & Enrollment Management, Batch & Campus Scheduling, Hostel & Logistics, LMS & Assessments, Verifiable Credentials, Career Counseling, and Recruitment Matching¹.
- **Intelligence & Natural Language Tier:** Bhashini ULCA REST and gRPC endpoints provide Neural Machine Translation (NMT), Automated Speech Recognition (ASR), and Text-to-Speech (TTS) capabilities¹⁷. Career advisory queries execute through a

LangChain-orchestrated Retrieval-Augmented Generation (RAG) pipeline backed by pgvector.

- **Persistence & Storage Tier:** Relational entities and vector embeddings are stored within PostgreSQL 16 with the pgvector extension¹. Ephemeral data, including session tokens and dynamic TOTP attendance secrets, reside in a high-availability Redis cluster¹. Binary media files, SCORM packages, and compiled PDF credentials are stored in S3-compatible, distributed object storage fronted by an edge Content Delivery Network (CDN)¹.

Detailed Functional Requirements

FR-01: Programme Catalogue, Online Registration & Society

Nomination Management

- **FR-01.1:** The system shall provide an administrative interface allowing authorized coordinators across VAMNICOM, RICMs, and ICMs to create, schedule, publish, edit, and archive training programs¹.
- **FR-01.2:** Each program entity shall capture mandatory metadata: unique course code, thematic cooperative domain (e.g., PACS Computerization, Dairy Management, Cooperative Banking, Cyber Security), target beneficiary profile, total instructional duration in hours, seat intake capacity, instructional languages, and eligibility criteria¹.
- **FR-01.3:** The system shall provide an open course discovery portal allowing rural youth and individual candidates to filter courses by sector, location, language, and mode, and submit digital self-enrollment applications¹.
- **FR-01.4:** Sponsoring cooperative bodies (PACS, DCCBs, State Cooperative Unions, SHG Federations) shall have access to an authenticated nomination portal to upload candidate rosters in bulk (CSV/XLSX) or submit individual nominations¹.
- **FR-01.5:** The registration engine shall automatically validate candidate eligibility rules and check historical records using unique civil identifiers (such as Aadhaar SHA-256 hashes or APAAR IDs) to flag duplicate enrollments across overlapping batches¹.
- **FR-01.6:** The system shall maintain an immutable, auditable workflow tracking each application through the following lifecycle states: SUBMITTED, SPONSOR_VERIFIED, UNDER_REVIEW, ACCEPTED, WAITLISTED, REJECTED, and ENROLLED¹.

FR-02: Unified Trainee, Trainer & Institutional Profile Management

- **FR-02.1:** The platform shall maintain a persistent Trainee Profile tracking demographic attributes, contact details, socio-economic classifications, educational history, completed NCCT courses, attendance records, competencies, and digital credentials¹.
- **FR-02.2:** Trainees shall have self-service access to update non-sensitive personal details (such as current address and mobile number), while primary civil identifiers and official training records remain read-only once administratively verified¹.
- **FR-02.3:** The platform shall maintain Institutional Profiles for NCCT headquarters, VAMNICOM, the 5 RICMs, 14 ICMs, and participating junior centers, detailing campus facilities, physical capacities, operational faculty, and accredited affiliations¹.

- **FR-02.4:** The system shall maintain Trainer/Faculty Profiles capturing academic qualifications, institutional affiliations, core subject competencies, historical session evaluations, and dynamic scheduling availability¹.

FR-03: Training Batch Lifecycle, Dynamic Timetable & Hostel/Logistics Management

- **FR-03.1:** The system shall organize accepted trainees into discrete batches, assigning designated batch coordinators, instructional dates, and training venues¹.
- **FR-03.2:** A scheduling engine shall generate conflict-free session timetables, mapping instructors, physical lecture halls, computer labs, and cohorts¹. The system shall flag and block scheduling conflicts before calendar publication¹.
- **FR-03.3:** Timetable updates, room reassignments, or cancellations shall trigger automated real-time alerts across mobile push notifications, SMS gateways, and in-app feeds for all affected trainees and faculty¹.
- **FR-03.4:** The system shall incorporate a Hostel Management module tracking residential room inventories, bed capacities, wing assignments, and maintenance status across residential campuses¹.
- **FR-03.5:** The hostel allocation engine shall automatically assign trainees to rooms upon batch finalization by solving an assignment problem designed to maximize room occupancy while respecting gender segregation rules, scheduling boundaries, and institutional quotas:

$$\min \sum_{i \in \mathcal{T}} \sum_{j \in \mathcal{R}} c_{ij} X_{ij}$$

subject to:

$$\sum_{j \in \mathcal{R}} X_{ij} = 1 \quad \forall i \in \mathcal{T}, \quad \sum_{i \in \mathcal{T}} X_{ij} \leq \text{Capacity}_j \quad \forall j \in \mathcal{R}, \quad X_{ij} \in \{0, 1\}$$

where $\mathcal{T}$ represents enrolled residential trainees, $\mathcal{R}$ denotes available rooms, Capacity_j

reflects bed limits, and c_{ij} penalizes sub-optimal cohort mixing.

- **FR-03.6:** The platform shall manage campus dining hall operations, recording meal attendance, tracking dietary preferences, and automating daily stipend and mess billing reconciliations for sponsoring societies¹.

FR-04: Digital Attendance Subsystem (Edge Biometric & Dynamic QR)

- **FR-04.1:** The platform shall support edge facial recognition attendance capture on COTS hardware (Android tablets, smartphones, and webcams)¹.
- **FR-04.2:** During initial onboarding, the client shall extract a 512-dimensional facial embedding vector using an edge-quantized deep convolutional neural network (MobileFaceNet with ArcFace margin loss):

$$\mathbf{e} \in \mathbb{R}^{512}, \quad \|\mathbf{e}\|_2 = 1$$

Raw facial images shall be purged from volatile memory immediately after embedding generation¹³.

- **FR-04.3:** The edge attendance terminal shall perform anti-spoofing liveness verification (evaluating micro-textures, randomized blink detection, and head-turn prompts) with an accuracy rate of $\geq$.
- **FR-04.4:** Verification shall compute the cosine similarity between the captured embedding and the locally cached template:

$$S_{\cos}(\mathbf{e}_{\text{live}}, \mathbf{e}_{\text{template}}) = \mathbf{e}_{\text{live}} \cdot \mathbf{e}_{\text{template}}$$

A successful attendance event shall be registered locally when $S_{\cos} \geq 0.68$, achieving a False Acceptance Rate (FAR) of $\leq$ and an edge execution time of $\leq 800 \text{ ms}$.

- **FR-04.5:** As an alternative to biometric scanning, the instructor's terminal shall support dynamic QR code attendance¹. The display shall render an HMAC-SHA256 signed Time-Based One-Time Password (TOTP) QR code that refreshes every 15 seconds:

$$\text{Payload} = \text{BatchID} \parallel \text{SessionID} \parallel T_k \parallel \text{HMAC}_K(\text{BatchID} \parallel \text{SessionID} \parallel T_k)$$

where $T_k = \lfloor t_{\text{epoch}} / 15 \rfloor$ and K represents the shared session secret.

- **FR-04.6:** The trainee's mobile app shall scan the dynamic QR code and validate both device clock synchronization and geographic positioning (using GPS/NavIC coordinates within a 50-meter geofenced radius of the venue) before recording the attendance entry.
- **FR-04.7:** When training sessions operate in offline environments, attendance records shall be cryptographically signed by the terminal, stored in an encrypted local buffer, and synchronized via an asynchronous queue once network access returns¹.

FR-05: Multilingual E-Learning & LMS Subsystem

- **FR-05.1:** The platform shall provide an authoring and management console for trainers to upload multimedia course content, including SCORM 2004 packages, HTML5 modules, PDF guides, and video lectures¹.
- **FR-05.2:** The LMS shall natively integrate with MeitY's Bhashini Universal Language Contribution APIs (ULCA) to provide automated translation of textual lesson materials across 22 Scheduled Indian Languages¹⁷.
- **FR-05.3:** The platform shall provide Bhashini-powered Text-to-Speech (TTS) audio narration for all lesson content, alongside Automated Speech Recognition (ASR) allowing trainees with limited digital literacy to interact using spoken Indic queries¹⁸.
- **FR-05.4:** The system shall track learner engagement, recording lesson completions, media playback positions, assessment attempts, and time-on-task metrics as structured

xAPI (Experience API) statements¹.

- **FR-05.5:** Trainees shall have the option to download selected learning modules to their mobile devices for offline study, maintaining full interactivity during periods without network connectivity¹.

FR-06: Interactive Assessment, Grading & Competency Tracking Engine

- **FR-06.1:** Instructors shall be able to create formative quizzes, practical assignments, and final summative examinations featuring multiple-choice, numerical entry, and structured text questions¹.
- **FR-06.2:** The assessment engine shall support configurable parameters, including question randomization, time limits, negative marking rules, and passing score thresholds (default: 75%)¹.
- **FR-06.3:** Formative assessments shall provide immediate automated grading and item-level feedback, highlighting incorrect answers and linking directly to remedial course modules¹.
- **FR-06.4:** Final examination attempts shall record proctoring telemetry, including focus-loss detection, window switching counts, and randomized face verification prompts.
- **FR-06.5:** Assessment scores and competency metrics shall map directly to National Occupational Standards (NOS) codes, updating the learner's central skill profile upon successful evaluation¹.

FR-07: Verifiable Skill Credentials, Certificate Repository & DigiLocker Issuance

- **FR-07.1:** The system shall automatically generate digital certificates once a trainee satisfies both program requirements: minimum attendance of 80% and a passing assessment score of $\geq 75\%$ ¹.
- **FR-07.2:** Certificates shall comply with the W3C Verifiable Credentials Data Model 2.0, formatted as JSON-LD documents signed using an Ed25519 cryptographic keypair held within NCCT's cloud Hardware Security Module (HSM) (Ed25519Signature2020)¹.
- **FR-07.3:** The credential engine shall integrate with DigiLocker and API Setu by hosting standardized Pull URI and Push URI callback endpoints²:
 - The Push URI endpoint shall register newly issued credential metadata with the central directory⁹.
 - The Pull URI endpoint shall dynamically verify and deliver the cryptographically signed XML/PDF certificate to a citizen's DigiLocker wallet upon request⁹.
- **FR-07.4:** Printed and PDF certificates shall feature a high-density, cryptographically signed 2D QR code¹. Scanning the QR code shall query the public verification endpoint, confirming credential authenticity and revocation status without exposing sensitive personal identifiers¹.
- **FR-07.5:** Authorized administrators shall have permissions to revoke issued certificates

for cause, automatically updating public revocation lists and notifying connected DigiLocker services¹.

FR-08: Indic Conversational Career Guidance & AI Counseling Chatbot

- **FR-08.1:** The system shall deploy an Indic conversational agent accessible via text and voice inputs across mobile and web interfaces¹.
- **FR-08.2:** Spoken Indic voice inputs shall route through Bhashini ASR pipelines to transcribe user queries, and generated responses shall synthesize into natural Indic speech via Bhashini TTS¹⁸.
- **FR-08.3:** The chatbot shall operate via Retrieval-Augmented Generation (RAG). User queries shall be converted into dense vector embeddings and matched against curated vector collections containing:
 - NCCT course catalogs and training pathways³.
 - Model Bye-Laws for Multi-Purpose PACS¹⁰.
 - PACS HR Policy frameworks and recruitment qualifications¹⁰.
 - Standard operating procedures for dairy, fisheries, and banking cooperatives⁶.
- **FR-08.4:** The RAG pipeline shall restrict responses to retrieved source contexts to eliminate speculative or inaccurate outputs, delivering grounded advice regarding cooperative career pathways, certification requirements, and relevant schemes¹.
- **FR-08.5:** The system shall incorporate guardrails to prevent the disclosure of confidential trainee records and include clear disclosures that chatbot recommendations do not guarantee employment¹.

FR-09: Cooperative Employer Dashboard, Vacancy Lifecycle & Talent Discovery

- **FR-09.1:** Verified cooperative employers (PACS, DCCBs, State Cooperative Banks, Milk Unions, FPOs) shall have access to a dedicated recruitment portal¹.
- **FR-09.2:** Employers shall be able to create, publish, edit, and close job postings, defining job roles, geographic work locations, educational prerequisites, minimum experience levels, and required NCCT skill certifications¹.
- **FR-09.3:** Employers shall be able to search the candidate directory using structured filters (such as district, course completed, assessment grade, and language proficiency)¹.
- **FR-09.4:** The candidate view shall display verified competencies and certificate credentials, while keeping contact information private until the candidate formally accepts an interview invitation¹.
- **FR-09.5:** The portal shall provide recruitment pipeline management tools, allowing employers to track candidates through stages: APPLIED, SHORTLISTED, INTERVIEW_SCHEDULED, OFFERED, PLACED, and REJECTED¹.

FR-10: AI-Assisted Candidate-Job Semantic Matching Engine

- **FR-10.1:** The platform shall compute a deterministic match score $S(c, j) \in [0, 1]$

between candidate profile c and job posting j using a multi-factor ranking function:

$$S(c, j) = w_{\text{skill}} S_{\text{semantic}}(\mathbf{v}_c, \mathbf{v}_j) + w_{\text{cert}} C(c, j) + w_{\text{exp}} E(c, j) + w_{\text{geo}} G(c, j)$$

where normalized weights satisfy:

$$w_{\text{skill}} + w_{\text{cert}} + w_{\text{exp}} + w_{\text{geo}} = 1.0$$

(Default configuration: $w_{\text{skill}} = 0.35, w_{\text{cert}} = 0.30, w_{\text{exp}} = 0.15, w_{\text{geo}} = 0.20$).

- **FR-10.2:** The semantic similarity component $S_{\text{semantic}}(\mathbf{v}_c, \mathbf{v}_j)$ shall compute the cosine similarity between the candidate's skill vector and the job requirement vector within a 512-dimensional embedding space.
- **FR-10.3:** The certification match factor $C(c, j)$ shall equal the proportion of mandatory NCCT certifications verified in the candidate's profile¹.
- **FR-10.4:** The experience factor $E(c, j)$ shall evaluate documented domain experience against job criteria, normalized over a configurable scale.
- **FR-10.5:** The geographic proximity factor $G(c, j)$ shall apply an exponential decay over the distance $d(c, j)$ in kilometers between the candidate's residence and the cooperative society:

$$G(c, j) = \exp(-\lambda \cdot d(c, j))$$

where $\lambda = 0.02$, prioritizing local rural youth for community society openings.

- **FR-10.6:** The matching engine shall generate transparent match explanations, outlining the specific certifications, skills, and geographic proximity factors contributing to a candidate's ranking¹.
- **FR-10.7:** Final recruitment and hiring decisions shall remain under human managerial authority, in compliance with ethical AI principles¹.

FR-11: Data Analytics, Executive MIS & National Outreach Dashboards

- **FR-11.1:** The platform shall provide role-tailored dashboards displaying enrollment, attendance, completion, assessment pass rates, and placement figures in real time¹.
- **FR-11.2:** Institutional Directors (VAMNICOM, RICMs, ICMs) shall have access to operational dashboards showing campus capacity utilization, hostel occupancy, faculty workloads, and batch timelines¹.
- **FR-11.3:** Central Ministry and NCCT administrators shall have access to a national MIS portal providing cross-state comparisons, sectoral skill gap analyses, and monitoring of national training goals (such as PACS computerization progress)¹.

- **FR-11.4:** Analytics reports shall support export to standard formats (CSV, XLSX, PDF) and include parameter filters for state, district, institution, date ranges, and demographic categories¹.
- **FR-11.5:** The system shall track and identify training dropouts and dormant regional societies to support future institutional outreach programs¹.

FR-12: Offline-First Mobile Synchronization Architecture

- **FR-12.1:** The mobile application and edge clients shall operate fully during intermittent or extended wide-area network (WAN) outages of up to 14 consecutive days¹.
- **FR-12.2:** Local operations—including attendance scanning, video playback, reading progress, and quiz completions—shall persist inside an encrypted SQLite database using Write-Ahead Logging (WAL)¹.
- **FR-12.3:** Upon network reconnection, the client shall initiate a delta synchronization handshake with the central server, transmitting encrypted JSON-formatted transaction batches¹.
- **FR-12.4:** State synchronization shall apply Conflict-Free Replicated Data Types (CRDTs) and Lamport Timestamps, resolving updates monotonically:

$$State_{server} \leftarrow \max(State_{server}, State_{client})$$

This protocol prevents progress rollbacks and prevents newer offline progress logs from being overwritten by older server records.

- **FR-12.5:** Synchronization failures shall trigger exponential backoff retry cycles, logging errors to an audit table without disrupting active client sessions¹.

FR-13: Multichannel Notification & Communication Gateway

- **FR-13.1:** The platform shall incorporate an event-driven notification engine that automatically triggers alerts based on system milestones¹.
- **FR-13.2:** Supported notification channels shall include Firebase Cloud Messaging (FCM) push notifications, SMS gateways, and automated emails¹.
- **FR-13.3:** The system shall issue notifications for registration confirmations, batch scheduling, timetable modifications, hostel allocations, assessment openings, certificate publications, and interview invitations¹.
- **FR-13.4:** Users shall have self-service preferences to manage notification channels for non-essential communications while essential administrative alerts remain mandatory¹.

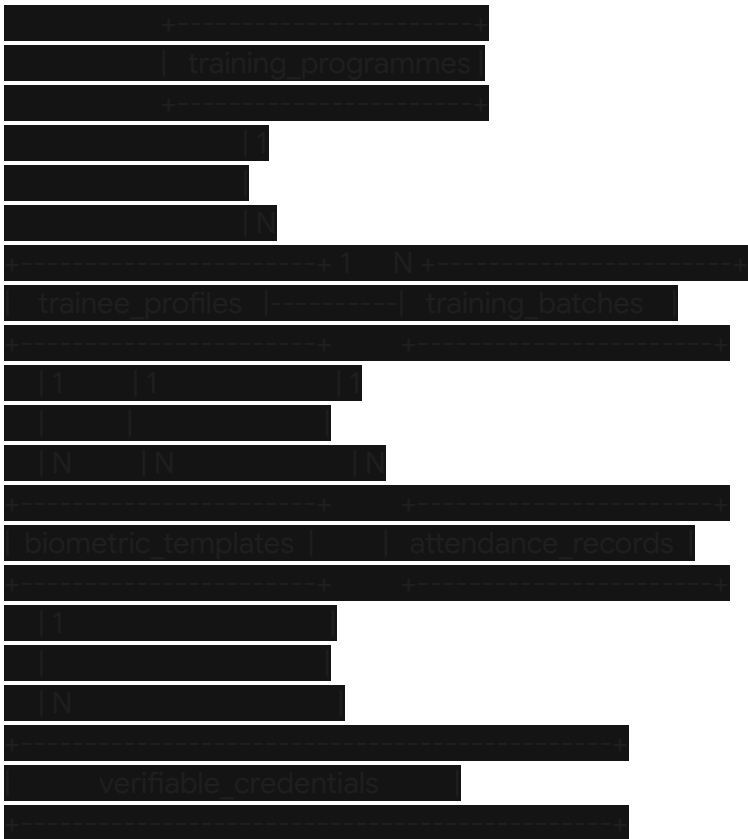
FR-14: Administrative Governance, Role-Based Access Control & Immutable Auditing

- **FR-14.1:** The system shall enforce fine-grained Role-Based Access Control (RBAC), validating user roles and permissions on every incoming API request¹.
- **FR-14.2:** Critical operations—such as manual attendance overrides, credential revocations, user role changes, and hostel room reallocations—shall require secondary confirmation and documented justification¹.
- **FR-14.3:** The platform shall maintain an append-only audit log table recording user ID,

action type, entity ID, client IP, payload SHA-256 hash, and timestamp for all system events¹.

- **FR-14.4:** Audit logs shall be cryptographically chained and protected from modification or truncation by regular administrative accounts¹.

Data Dictionary and Persistent Schema Specifications



Core Database Entities

Entity Table Name	Primary Key	Key Attributes and Types	Constraints and Relational Indexes
trainee_profiles [cite: 1]	trainee_id (UUID)	user_id (UUID), aadhaar_sha256 (VARCHAR(64)), full_name	Unique on aadhaar_sha256; composite B-Tree index on (state_id,

		(VARCHAR(128)), dob (DATE), gender (VARCHAR(16)), district_id (INT), state_id (INT), education_level (VARCHAR(64)), created_at (TIMESTAMPTZ).	district_id).
biometric_templates [cite:]	template_id (UUID)	trainee_id (UUID), embedding_vector (VECTOR(512)), algorithm_tag (VARCHAR(32)), liveness_score (FLOAT), updated_at (TIMESTAMPTZ).	FK to trainee_profiles; HNSW index on embedding_vector using cosine distance metric.
training_programmes [cite: 1]	programme_id (UUID)	institution_id (UUID), course_code (VARCHAR(32)), title (VARCHAR(256)), sector_code (VARCHAR(32)), duration_hours (INT), intake_limit (INT), status (VARCHAR(16)).	Unique on course_code; B-Tree index on (institution_id, sector_code).
training_batches [cite: 1]	batch_id (UUID)	programme_id (UUID), lead_trainer_id (UUID), start_date (DATE), end_date (DATE), facility_id (UUID), batch_state (VARCHAR(32)).	Check constraint (end_date >= start_date); B-Tree index on batch_state.

attendance_records [cite: 1]	record_id (UUID)	batch_id (UUID), trainee_id (UUID), session_id (UUID), recorded_at (TIMESTAMPTZ), method (VARCHAR(16)), similarity_score (FLOAT), device_id (VARCHAR(64)).	Composite index on (batch_id, session_id, trainee_id); partitioned monthly.
verifiable_credentials [cite:]	credential_id (UUID)	trainee_id (UUID), batch_id (UUID), certificate_urn (VARCHAR(128)), issued_at (TIMESTAMPTZ), signature_hex (TEXT), digilocker_uri (VARCHAR(256)), status (VARCHAR(16)) ⁹ .	Unique on certificate_urn; B-Tree index on digilocker_uri.
job_postings [cite: 1]	job_id (UUID)	recruiter_id (UUID), society_type (VARCHAR(64)), title (VARCHAR(128)), skills_array (TEXT[]), district_id (INT), vacancies (INT), requirement_vector (VECTOR(512)).	HNSW vector index on requirement_vector ; GIN index on skills_array.
audit_ledger_events [cite: 1]	event_id (UUID)	actor_id (UUID), action_type (VARCHAR(64)), target_entity_id (UUID), client_ip (VARCHAR(45)),	Immutable table configuration; append-only rules; B-Tree index on (timestamp, actor_id).

		payload_hash (VARCHAR(64)), timestamp (TIMESTAMPTZ).	
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Data Exchange Formats and Contract Specifications

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|    "programmeTitle": "PACS Computerization and Common Accounting System"
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    "language": {"sourceLanguage": "mr"}
    "gender": "female"
  }
  "inputData":
  {
    "audio": [{"audioContent": "UkiGRiQAAABXOVZFZm10IBAAAAAABAAEA..."}]
  }
}
```

```
{
  "syncBatchId": "c4b9281a-6d1b-48ae-94a2-19273c5bb201"
  "edgeDeviceId": "KIOSK-ICM-AHM-002"
  "batchId": "BATCH-2026-OCT-094"
  "syncTimestamp": "2026-10-07T18:30:00Z"
  "recordsCount": 42
  "records":
  {
    "traineeId": "a1b2c3d4-e5f6-7a8b-9c0d-1e2f3a4b5c6d"
    "sessionId": "s7t8u/v6-w5x4-3y2z-1a0b-cdef01234567"
  }
}
```

```
{
  "timestamp": "2026-10-07T09:15:22",
  "method": "FACE_RECOGNITION",
  "similarityScore": 0.892,
  "livenessScore": 0.994,
  "terminalSignature": "MEQCIF6..AIBe5="
}
```

RESTful API Endpoints Specification

API Group / Path	Method	Ingress Payload / Parameters	Egress Response Schema	Security Scope
/api/v1/programmes [cite: 1]	POST	title, sector_code, duration_hours, capacity, languages ¹ .	201 Created: Full program object with assigned UUID ¹ .	role:INSTITUTION_ADMIN [cite: 1]
/api/v1/registrations/nominate [cite: 1]	POST	programme_id, array of candidate profiles, society registration ID ¹ .	200 OK: Nomination summary, accepted count, waitlist count ¹ .	role:SOCIETY_ADMIN [cite: 1]
/api/v1/batches/{id}/timetable [cite: 1]	GET	batch_id path parameter; optional date range query ¹ .	200 OK: Array of scheduled sessions with trainer and venue ¹ .	role:TRAINEE, TRAINER [cite: 1]
/api/v1/attendance/verify-qr [cite: 1]	POST	totp_token, device_lat, device_lng, scanned_timestamp.	200 OK: Attendance confirmation record with status MARKED ¹ .	role:TRAINEE [cite: 1]

/api/v1/attendance/edge-sync [cite: 1]	POST	Signed batch of attendance events from edge terminal ¹ .	200 OK: Sync confirmation with processed and rejected counts ¹ .	mTLS / role:EDGE_DEVICE [cite: 1]
/api/v1/lms/courses/{id}/sync [cite: 1]	POST	Client CRDT delta array of xAPI learning event records.	200 OK: Merged course state, completion percentage, badge flags ¹ .	role:TRAINEE [cite: 1]
/api/v1/credentials/digilocker/pull [cite:]	POST	Standardized DigiLocker XML request containing citizen URI ⁹ .	200 OK: Base64 encoded signed XML certificate payload ⁹ .	DigiLocker HMAC Auth [cite:]
/api/v1/career/counsel [cite: 1]	POST	query_text or query_audio_base64, language_code ¹ .	200 OK: Synthesized audio stream and textual guidance response ¹ .	role:TRAINEE [cite: 1]
/api/v1/jobs/match [cite: 1]	POST	job_id, pagination parameters, district radius filter ¹ .	200 OK: Ranked array of matching candidates with score breakdown ¹ .	role:RECRUITER [cite: 1]

Hardware Configurations and Edge Deployment Model

In accordance with the Software + Hardware deployment model for Problem Statement ID 26087, the platform pairs its cloud services with an edge computing architecture¹. This design uses standard commercial off-the-shelf (COTS) devices to deliver identity verification and localized learning without requiring custom hardware manufacturing¹.



Deployment Hardware Role	Recommended Form Factor	Processing, Memory & Storage	Sensors & Peripherals	Institutional Deployment Environment
Campus Edge Micro-Server [cite: 1]	1U Rackmount Server or Small Form Factor (SFF) PC.	x86-64 8-Core Processor, 32 GB ECC RAM, 1 TB NVMe SSD Storage.	Dual Gigabit Ethernet (RJ-45), Hardware TPM 2.0 Module.	Server rooms at VAMNICOM, 5 RICMs, and 14 ICM campus hubs ¹ .
Classroom Biometric Kiosk [cite: 1]	Wall-mounted ruggedized tablet or industrial SBC enclosure.	ARM Octa-Core Processor, 4 GB RAM, 64 GB eMMC, $\geq$ TOPS	5MP RGB Camera, 2MP Near-Infrared (NIR) Camera, IR LED illuminator.	Lecture halls, IT laboratories, and seminar halls across campuses ¹ .

		NPU.		
Mobile Attendance Terminal [cite: 1]	Commercial Android Smartphone or Tablet (COTS) ¹ .	Quad-Core 2.0 GHz Processor, 3 GB RAM, Android 11+ OS.	8MP Autofocus Camera, NavIC / GPS Positioning Sensor.	Field trainers conducting outreach sessions at remote PACS centers ¹ .
Logistics & Mess Kiosk [cite: 1]	Desktop POS Terminal or Wall Enclosure.	Quad-Core ARM Processor, 2 GB RAM, 32 GB Storage.	High-speed 2D QR Scanner, Optional Mifare 13.56 MHz RFID Reader.	Campus hostel reception desks, dining halls, and campus security gates ¹ .

Non-Functional Requirements (NFR-01 to NFR-12)

Requirement ID	Performance Dimension	Quantitative Metric / Specification Target	Enforcement Mechanism and Architecture Strategy
NFR-01: Performance [cite: 1]	API Ingress Latency ¹	$\leq$ for 95th percentile requests; $\leq$ for 99th percentile under peak loads.	Read-replica database pools, Redis in-memory query caching, CDN edge delivery ¹ .
NFR-02: Scalability [cite: 1]	Concurrent Platform Capacity ¹	Support for 25,000 concurrent active learners; 2,500 attendance API transactions per second.	Horizontal Pod Autoscaling (HPA) on Kubernetes; stateless application design.
NFR-03: Availability [cite: 1]	Operational Uptime ¹	99.9% annual service availability (	Multi-Availability-Zone deployment, health checks,

		$\leq$ hours unplanned downtime per year).	automatic database failover.
NFR-04: Security [cite: 1]	Cryptographic Transport & Rest ¹	TLS 1.3 enforced for all transport links; AES-256-GCM encryption for persistent databases and S3 buckets.	Strict Transport Security (HSTS), automated HSM key rotation, OAuth 2.0 / OIDC tokens ¹ .
NFR-05: Privacy [cite: 1]	Biometric & Trainee Data Protection ¹	Full compliance with DPDP Act 2023; zero storage of raw biometric facial imagery ¹³ .	Immediate purging of raw camera frames; irreversible salted 512-dimensional vector embedding.
NFR-06: Usability [cite: 1]	User Experience Accessibility ¹	System System Usability Scale (SUS) score $\geq$; task completion rate $\geq$ for first-time rural learners.	Simple touch layouts, voice-guided Indic interactions, minimal text entry forms ¹ .
NFR-07: Accessibility [cite: 1]	Universal Digital Access ¹	Full compliance with WCAG 2.1 Level AA and Government of India Guidelines for Indian Government Websites (GIGW).	High-contrast visual modes, text-to-speech alternatives, keyboard navigation support.
NFR-08: Localization [cite: 1]	Multilingual Operational Coverage ¹	Complete interface localization and automated translation across	Integration with MeitY Bhashini ULCA translation pipelines and

		22 Scheduled Indian Languages ¹⁷ .	speech synthesis models ¹⁷ .
NFR-09: Maintainability [cite: 1]	Architectural Modularity ¹	Code test coverage $\geq$; automated CI/CD pipeline deployments with zero-downtime rolling upgrades.	Domain-driven microservice isolation, containerized testing suites, structured OpenAPI documentation.
NFR-10: Reliability [cite: 1]	Data Consistency & Integrity ¹	Zero transaction loss during network partitions; guaranteed ACID compliance for academic records ¹ .	Distributed SQLite delta sync with CRDT conflict-free monotonic merge operations ¹⁵ .
NFR-11: Auditability [cite: 1]	Governance Traceability ¹	100% of administrative modifications, attendance overrides, and credential actions logged with immutable hashes ¹ .	Cryptographically linked append-only audit tables; tamper-evident log streaming to secure SIEM ¹ .
NFR-12: Interoperability [cite: 1]	Digital Public Infrastructure Links ¹	Standardized integration with DigiLocker, API Setu, Skill India Digital Hub (SIDH), and National Career Service (NCS) ² .	Adherence to Open API standards, W3C Verifiable Credentials 2.0 schemas, and secure JSON-LD formatting ²⁰ .

Statutory Compliance, Security & Privacy Architecture

The platform's security and identity governance architecture aligns with the statutory mandates of the Digital Personal Data Protection (DPDP) Act, 2023, and the DPDP Rules, 2025¹³.

- Data Fiduciary Governance:** The National Council for Cooperative Training (NCCT) operates as the designated Data Fiduciary, maintaining formal governance over all data processing activities executed across VAMNICOM, RICMs, ICMs, and junior centers¹.

- **Purpose Limitation and Consent Collection:** In compliance with Section 6 of the DPDP Act, collected trainee information and biometric records are used strictly for training registration, attendance auditing, and certification, and cannot be repurposed for external surveillance or commercial marketing. The system presents a clear, concise notice in the user's chosen language, detailing the exact data elements collected, operational purposes, and retention timelines. Consent is recorded as an auditable, timestamped record. Trainees maintain the statutory right to withdraw biometric consent at any point, in which case the system defaults to dynamic QR code attendance verification¹.
- **Biometric Privacy and Template Minimization:** Raw facial images, video recordings, and unencrypted templates are never written to permanent disk storage or transmitted across network links. During enrollment, live facial images are captured in volatile memory, converted into 512-dimensional floating-point embedding vectors ($\mathbf{e} \in \mathbb{R}^{512}$), and purged immediately from RAM¹³. Embedding vectors are salted with tenant-specific keys and encrypted using AES-256-GCM before storage, preventing facial image reconstruction and mitigating identity theft risks.
- **Automated Data Retention Limits:** In accordance with Section 8(7) of the DPDP Act, biometric templates are removed from edge verification devices once a training batch concludes and permanently erased from central databases 90 days after course completion, unless explicit consent is renewed for ongoing certifications²⁴.
- **Incident Management and CERT-In Reporting:** The platform incorporates automated intrusion detection controls. Any unauthorized access attempts trigger automated isolation protocols and generate immutable audit logs, while notifying administrative officers and CERT-In within mandated reporting windows¹.

Core Operational Workflows

Trainee Journey Workflow

1. **Profile Registration:** The rural candidate accesses the web portal or mobile app, enters demographic details, and validates identity via mobile OTP¹.
2. **Program Enrollment:** The candidate discovers available training programs, reviews eligibility criteria, and submits an online registration or receives an organizational nomination from their PACS/society¹.
3. **Batch Allocation & Check-in:** Upon approval, the candidate receives batch, timetable, and hostel room assignments¹.
4. **Attendance Marking:** During sessions, the trainee marks attendance through classroom facial recognition or by scanning dynamic QR codes via the mobile app¹.
5. **Instruction & Offline Study:** The candidate accesses multilingual learning modules, downloading lesson content for offline access when necessary¹.
6. **Assessment & Completion:** The trainee completes module quizzes and the final certification exam, with engagement tracked via xAPI statements¹.
7. **Credential Issuance:** Upon satisfying the 80% attendance and 75% score thresholds, a

signed W3C Verifiable Credential is generated and pushed to the trainee's DigiLocker wallet¹.

8. **Career Guidance & Job Application:** The trainee consults the Indic AI career chatbot, reviews recommended cooperative job openings, and submits applications with verified credentials¹.

Institutional Administrator Workflow

1. **Course Configuration:** The administrator publishes upcoming courses in accordance with the Annual Training Calendar¹.
2. **Nomination Processing:** The coordinator reviews candidate submissions from sponsoring cooperative federations, verifies eligibility, and approves batch rosters¹.
3. **Resource Scheduling:** The administrator schedules training facilities, IT labs, and faculty assignments, with the engine resolving calendar conflicts¹.
4. **Logistics Allotment:** The system runs the hostel allocation engine to assign residential quarters and sends projected meal counts to dining facilities¹.
5. **Attendance Oversight:** The administrator monitors attendance dashboards in real time, auditing automated edge sync batches and managing justified attendance exceptions¹.
6. **Certification Approval:** The coordinator approves batch completion records, triggering automated cryptographic signing and DigiLocker credential issuance¹.
7. **Analytics Review:** The administrator exports program outcome reports for institutional reporting to NCCT headquarters¹.

Cooperative Employer & Recruiter Workflow

1. **Employer Onboarding:** The cooperative employer (such as a PACS, District Central Cooperative Bank, or Dairy Federation) registers using their official society registry identifier and undergoes administrator verification¹.
2. **Requisition Publication:** The recruiter publishes job openings, defining required competencies, minimum experience levels, and prerequisite NCCT course certifications¹.
3. **Talent Discovery:** The recruiter uses candidate search filters or runs the AI matching engine to generate a ranked candidate list sorted by semantic skill alignment and geographic proximity¹.
4. **Credential Verification:** The employer scans candidate credential QR codes or checks the public verification endpoint to validate certificate authenticity directly against the issuer registry¹.
5. **Hiring Pipeline Tracking:** The recruiter shortlists candidates, issues interview invitations via system notification channels, and records final hiring outcomes in the recruitment module¹.

SIH MVP Implementation vs Full Production Roadmap

SIH Hackathon MVP Deliverables

To demonstrate the platform's core architecture during the competition, the prototype focuses on the following primary functional capabilities¹:

- Trainee registration, profile management, and program enrollment workflows¹.

- Training batch calendar and interactive timetable dashboard¹.
- Functional QR-based session attendance, alongside a working demonstration of on-device face verification using a standard webcam or Android smartphone¹.
- Multilingual LMS interface featuring sample training modules on PACS Computerization and Common Accounting System (CAS) frameworks¹.
- Formative assessment module with automated scoring and completion tracking¹.
- Cryptographically signed digital certificate generation featuring scannable verification QR codes¹.
- Indic conversational career counseling chatbot running grounded RAG queries against cooperative training documentation¹.
- Recruiter dashboard featuring job posting workflows and semantic candidate ranking¹.
- Management analytics view displaying program, attendance, and certification metrics¹.
- Demonstration of offline-capable mobile UI operation with local transaction buffering and simulated synchronization¹.

Production Implementation Roadmap

Phase Window	Primary Milestones and Engineering Scope	Institutional Participants	Completion Gate Targets
Phase 1: Enterprise ERP Core (Months 1 – 6) [cite:]	Deployment of multi-tenant cloud ERP core; online nomination engine; dynamic QR attendance; hostel inventory management ¹ .	NCCT HQ, VAMNICOM Pune, 5 RICMs, 14 ICMs ¹ .	100% paperless course nomination; automated batch scheduling operational across all 20 campuses ¹ .
Phase 2: Edge Biometrics & Multilingual LMS (Months 7 – 12) [cite:]	Installation of classroom biometric kiosks; launch of offline-first mobile LMS; integration with Bhashini ULCA pipelines and DigiLocker API Setu ¹ .	20 Primary Training Campuses; Pilot PACS Batches ¹ .	On-device face recognition operational; verifiable credentials active in citizen DigiLocker accounts ¹ .

Phase 3: AI Ecosystem & Employment Exchange (Months 13 – 18) [cite:]	Rollout of voice-enabled Indic career chatbot; launch of cooperative recruitment exchange; bidirectional data sync with NCS and SIDH; deployment to junior training centers ¹ .	All 20 Institutes, ~109 JCTCs, Cooperative Societies & Banks ¹ .	Full semantic candidate matching active; documented employment placements across cooperative societies ¹ .
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Verification, Testing Strategy, and Acceptance Criteria

Comprehensive Quality Assurance Matrix

Quality Test Domain	Target Subsystem Under Test	Testing Methodology and Tools	Certified Acceptance Threshold
Biometric Verification [cite: 1]	Edge Facial Kiosk & MobileFaceNet Engine.	Benchmark evaluation against 10,000 diverse facial test samples across varying lighting environments.	False Acceptance Rate $\leq$; False Rejection Rate $\leq$; Presentation Attack Detection $\geq$.
Offline Synchronization [cite: 1]	Mobile Client, SQLite/Room Database, Sync Gateway ¹ .	72-hour simulated network partition during which 500 local attendance and learning events are logged, followed by reconnection.	Zero record loss; zero data corruption; all client transactions reconciled within 60 seconds ¹ .
Credential Integrity	Credential Signer, DigiLocker	Automated issuance, signature	100% compliance with W3C VC 2.0

[cite:]	Pull/Push Callbacks ⁹ .	validation, schema parsing, and high-frequency callback tests ⁹ .	schema; public cryptographic key verification success rate = .
AI Recommendation Grounding [cite: 1]	Indic RAG Career Chatbot ¹ .	Evaluation against 1,000 synthetic career queries across Hindi, Marathi, Tamil, and English.	Zero hallucinated policy outputs; context-retrieval relevance score $\geq$; response latency $\leq$.
Load and Stress [cite: 1]	Ingress API Gateway, Core Application Mesh, Database ¹ .	Distributed stress testing simulating 25,000 concurrent user sessions and 2,500 simultaneous attendance submissions.	95th percentile latency $\leq$; zero unhandled HTTP 5xx errors; server CPU utilization $\leq$.

User Acceptance Criteria

- An individual or sponsoring society can register, complete enrollment, and view the trainee journey through the portal¹.
- An administrator can create a program, generate dynamic timetables, and allocate hostel accommodations without scheduling overlaps¹.
- Daily attendance records can be captured via dynamic QR code or on-device edge facial recognition¹.
- A trainee can access interactive learning materials, take assessments, and view completion status¹.
- The system validates completion criteria and issues a tamper-evident, verifiable digital credential accessible via DigiLocker¹.
- The conversational career chatbot answers Indic voice and text queries using grounded training documentation¹.
- A cooperative employer can publish job vacancies and view candidate recommendations ranked by verified skills and geographic proximity¹.
- The mobile application maintains core learning workflows during network outages and synchronizes progress upon reconnection¹.
- Role-based access control restricts unauthorized users from viewing or modifying protected data¹.

Requirement Traceability Matrix and Risk

Management

Requirement Traceability Matrix

Problem Statement Feature	SRS Functional Requirement Coverage	Verification Method
Online programme registration and nomination management [cite: 1]	FR-01: Programme Catalogue, Online Registration & Society Nomination Management ¹ .	Functional Workflow Testing ¹ .
Participant, institution, and trainee profile management [cite: 1]	FR-02: Unified Trainee, Trainer & Institutional Profile Management ¹ .	Integration & Security Testing ¹ .
Interactive multilingual e-learning modules [cite: 1]	FR-05: Multilingual E-Learning & LMS Subsystem ¹ .	Usability & Localization Auditing ¹ .
Digital attendance through face recognition / QR code [cite: 1]	FR-04: Digital Attendance Subsystem (Edge Biometric & Dynamic QR) ¹ .	Biometric & Edge Hardware Benchmarking ¹ .
Timetable, hostel, and logistics management [cite: 1]	FR-03: Training Batch Lifecycle, Dynamic Timetable & Hostel/Logistics Management ¹ .	Algorithmic Optimization & Stress Tests ¹ .
LMS integration with assessments and certification [cite: 1]	FR-06: Interactive Assessment, Grading & Competency Tracking Engine ¹ .	Automated Unit & Integration Testing ¹ .
Skill certification repository and verification [cite: 1]	FR-07: Verifiable Skill Credentials, Certificate Repository & DigiLocker Issuance ¹ .	Cryptographic & Schema Validation ¹ .

Career counseling chatbot [cite: 1]	FR-08: Indic Conversational Career Guidance & AI Counseling Chatbot ¹ .	RAG Grounding & Conversational Evaluation ¹ .
Employer and recruiter dashboard [cite: 1]	FR-09: Cooperative Employer Dashboard, Vacancy Lifecycle & Talent Discovery ¹ .	User Interface & Security Role Testing ¹ .
AI-Assisted Job Matching [cite: 1]	FR-10: AI-Assisted Candidate-Job Semantic Matching Engine ¹ .	Algorithmic Accuracy & Bias Testing ¹ .
Data analytics & centralized database [cite: 1]	FR-11: Data Analytics, Executive MIS & National Outreach Dashboards ¹ .	Data Aggregation & Query Plan Testing ¹ .
Mobile-friendly and offline-accessible learning platform [cite: 1]	FR-12: Offline-First Mobile Synchronization Architecture ¹ .	WAN Interruption & CRDT State Testing ¹ .

Comprehensive Risk Register and Mitigation Strategy

Risk Category & ID	Identified Operational Risk	Potential Impact	Severity & Probability	Mitigation and Architectural Strategy
Technical: R-01 [cite: 1]	Intermittent or absent internet connectivity at remote training locations ¹ .	Interrupted attendance verification and learning delivery ¹ .	High / High ¹	Offline-first mobile design; local SQLite caching; monotonic CRDT state synchronization upon reconnection ¹ .
Technical: R-02 [cite: 1]	High False Rejection Rates (FRR) in	Classroom delays; participant	High / Medium	Use of dual RGB+NIR cameras with

	edge facial recognition under challenging lighting ¹ .	frustration; inaccurate logs ¹ .		infrared illumination; adaptive thresholds; fallback to dynamic QR code scanning ¹ .
Security: R-03 [cite: 1]	Unauthorized exposure of trainee biometric or personal data ¹ .	Privacy violation; regulatory penalties under DPDP Act 2023 ¹ .	Critical / Low	Strict avoidance of raw image persistence; salted 512-dimensional vector hashing; AES-256 encryption at rest; strict RBAC ¹ .
AI Safety: R-04 [cite: 1]	Inaccurate advice or hallucinations generated by the career counseling chatbot ¹ .	Misleading guidance provided to rural jobseekers ¹ .	Medium / Medium	Strict RAG architectural boundaries; limiting generation to retrieved institutional policies; clear guidance disclaimers ¹ .
Operational: R-05 [cite: 1]	Algorithmic bias in recruitment candidate matching ¹ .	Unfair candidate rankings; exclusion of rural youth ¹ .	High / Low	Deterministic multi-factor scoring; transparent ranking criteria; human authority over all hiring decisions ¹ .

Infrastructure : R-06 [cite: 1]	High bandwidth consumption and storage costs from multimedia streaming ¹ .	Infrastructure budget overruns; slow media playback ¹ .	Medium / Medium	Transcoding to lightweight WebM/Opus formats; campus edge server micro-caching; distributed CDN asset delivery ¹ .
Governance: R-07 [cite: 1]	Low digital adoption among grassroots cooperative personnel ¹ .	Continued reliance on manual, paper-based workflows ¹ .	High / Medium	Simple, intuitive user interfaces; Indic voice navigation via Bhashini; phased transition support across institutes ¹ .

This comprehensive specification provides a standard-compliant engineering baseline for deploying an integrated training, capacity-building, and employment platform across NCCT institutions and India's cooperative sector¹.

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